ANABIONTICS

A Material Theory of Biological Persistence

Essay I

Hiram Dunn
AstroBiota
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1. Persistence as a Material Problem

Biology has an extraordinary language for parts. We can identify genes, proteins, metabolites, receptors, membranes, organelles, cells, organisms, and ecological partners with remarkable precision. What we describe less easily is how these components become a biological state that persists through time.

Living systems are not inventories. They have histories. They construct environments, maintain interfaces, recover from disturbance, transition between developmental states, and inherit material conditions produced by previous biological activity. Their individual components may turn over while the organization of the system remains recognizable.

Persistence therefore has a material biology of its own.

I call that domain anabiontics: the study of materials and material environments through which living systems establish, maintain, recover, transmit, and reconstruct biological state.

An anabiontic may be a small molecule, polymer, membrane, extracellular matrix, secretion, conditioned substrate, composite biological material, or constructed environment. The category is not defined primarily by chemistry. It is defined by the role that material occupies relative to biological state.

Anabiontics does not include every material that changes biology. Toxins change biological state. Antibiotics change biological state. Hormones, nutrients, injury, radiation, and drugs can all produce transitions. Transformation alone is not enough. The anabiontic question is more specific: what material participates in establishing, maintaining, recovering, transmitting, or reconstructing the state of a defined biological system?

2. Anabiont and Catabiont

Anabiontics requires a distinction between identity and role. Biology often describes coupled systems by naming their participants: host and symbiont, mycobiont and photobiont, cell and organelle. These terms identify actors, but identity alone does not tell us how a system organizes continuity and throughput.

The catabiont is the role disproportionately associated with throughput: resource acquisition, energetic conversion, production, expansion, turnover, and generative activity. The anabiont is the role disproportionately associated with persistence: buffering disturbance, maintaining interfaces, regulating difficult coupling, preserving continuity, repairing damage, and reconstructing state.

These are not taxonomic categories. An organism is not inherently an anabiont or a catabiont. The polarity exists only relative to a named coupled system and its constraints. The same actor may govern persistence at one scale while participating in throughput at another.

This is why anabiont and catabiont are broader than host and symbiont. The polarity can appear between organisms, but also between populations, cell types, tissues, organelles, developmental programs, and constructed biological environments. The object of classification is not the thing in isolation. It is the role the thing occupies in maintaining a particular state.

Anabionts govern persistence. Catabionts govern throughput.

3. Matter, Role, and State

Once the persistence role has been identified, the next question is material: what physically makes that role possible?

Molecular identity alone is insufficient. Two systems can contain similar components and nevertheless occupy radically different states because their spatial organization, interfaces, gradients, polymer architectures, developmental histories, and previous exposures differ. A molecule tells us what material is present. It does not, by itself, tell us what role that material occupies in the history of the system.

The same molecule may participate in persistence in one system and disruption in another. The same polymer may stabilize one ecology and destabilize another. Material identity can remain constant while biological role changes completely.

Matter has identity. Biological role is relational.

Anabiontics therefore describes a relationship between matter and state. It asks how material becomes architecture, how architecture becomes constraint, and how constraint changes the states that a living system can maintain or reconstruct.

This becomes especially apparent with polymers. Bacteria produce extracellular polymeric matrices. Fungi construct hyphal and extracellular architectures. Plants build walls and mucilage. Animals construct extracellular matrices. Spiders produce silk. These materials are often called structural, but structure is biologically active because structure determines what can happen next.

A persistent material can change diffusion, hydration, attachment, mechanical force, signaling, transport, protection, spatial organization, and future growth. Material produced during one biological state therefore becomes part of the constraint environment encountered by the next.

Biological state changes matter. Changed matter alters future constraint. Constraint shapes subsequent biological state.

A polymer can become history with geometry.

4. Living Architectures

A spider web illustrates the principle well. Initially, the spider produces silk. Once the web exists, however, the web begins shaping the spider. It determines where movement occurs, how vibration propagates, where prey is intercepted, where repair becomes necessary, and where future silk is deposited. Biological output has become biological environment.

The same principle appears even more clearly in biofilms. A biofilm constructs an extracellular environment that regulates hydration, diffusion, nutrient movement, adhesion, protection, signaling, and spatial organization. Material produced by previous biological activity becomes the environment experienced by later biological activity. The ecology begins to inherit itself materially.

Lichen pushes the idea further. A mature lichen cannot be adequately described as a fungus placed beside a photosynthetic partner. The thallus is an organized ecological state constructed through metabolism, hydration, substrate interaction, microbial association, material architecture, and environmental history. Destroy that organization and retaining the individual partners does not automatically preserve the lichen. The state must be reconstructed.

This is where anabiontics differs from simply cataloguing molecules. The relevant question is not only which compounds exist in the lichen, spider web, or biofilm. It is which materials and architectures allow those systems to remain historically continuous despite disturbance and turnover.

The same ontology extends inside organisms. Fungi transition between vegetative growth, reproductive development, sporulation, injury, recovery, and dormancy. Cells move among proliferative, differentiated, quiescent, and stressed states. Organelles reorganize as cellular conditions change. Tissues preserve identity despite enormous molecular turnover.

At every scale, biology confronts the same problem: how does an organized state remain continuous while the matter composing it changes?

5. Observation, Perturbation, and Persistence

This distinction changes how unusual material effects should be interpreted. In my own fungal cultures, psilocybin-containing media have been associated with a striking shift in growth architecture: translucent growth within and beneath the agar rather than the ordinary tomentose surface phenotype. The observation does not, by itself, establish mechanism. It establishes a state difference produced within a different material environment.

The anabiontic question is therefore not simply why psilocybin is psychedelic in mammals. It is what biological state the psilocybin chemical system may participate in establishing, maintaining, or reorganizing within the fungus that produces it. A molecule that profoundly disrupts mammalian state could simultaneously participate in persistence or state organization within a fungal system. There is no contradiction. The molecule is the same; the biological system is different.

Antibiotics reveal the same distinction from another direction. Antibiotic science asks which organism a molecule suppresses, through what mechanism, and with what resistance profile. Anabiontics asks what happens to the resulting state after the perturbation.

An antibiotic may reorganize a microbial ecology and then disappear. Some organisms decline, others expand, metabolic relationships shift, interfaces are reconstructed, and extracellular structures change. The molecule can be gone while the reorganized ecology remains.

Perturbation asks how State A can be disrupted. Anabiontics asks how State B becomes self-maintaining.

This suggests a different logic for biological engineering. Many difficult problems are approached by repeatedly forcing systems away from an unwanted state: kill the organism, inhibit the pathway, block the receptor, suppress the response. But if the underlying state continually reconstructs itself, stronger perturbation may never solve the persistence problem.

Anabiontic engineering would instead ask what material environment allows a desired state to establish itself, transfer persistence into the biological system, and resist displacement. The objective changes from repeatedly forcing biology toward a state to constructing the conditions through which that state can maintain or rebuild itself.

6. History as Biological Memory

Material persistence and state persistence are not the same quantity. A molecule can disappear rapidly while the state associated with its presence remains. Conversely, a molecule can remain for a long time without producing durable state change.

History therefore becomes part of the system. The first exposure to a material encounters one state. If that encounter changes the system, an identical second exposure encounters something different. The molecule can be identical. The concentration can be identical. The genome can be identical. The biological event is still different because the receiving system carries the trace of its previous state.

This is where my work with biophotograms becomes important. A biological system can encounter one environmental condition and later preserve that encounter as persistent phenotype or material trace. The original condition can disappear while the consequence remains readable.

A biophotogram is therefore useful not simply as an image of growth, but as a record of state. It demonstrates the distinction between what exists now and what the system has previously experienced.

Biological memory does not require symbolic representation. A spider web does not need to contain a description of the spider that constructed it. A biofilm matrix does not need to contain a symbolic account of its microbial community. A lichen thallus does not need to know its developmental history. The past persists because it changed the physical conditions of the future.

Previous state changes matter. Matter changes constraint. Constraint changes subsequent state. The resulting state changes matter again.

That loop is a fundamental form of biological memory.

7. From Persistence to Accessibility

Anabiontics begins with persistence, but persistence immediately raises a second problem. If material conditions help determine which biological states can be maintained, then those same conditions must also influence which states can be reached.

Anabiont and catabiont provide a role polarity for interpreting coupled systems. Anabiontics provides the material language for understanding how the persistence side of that polarity becomes physical. State Biology places both within a larger history: molecules participate in structures; structures create interfaces; interfaces constrain interactions; interactions establish states; states modify their environments; and those modified environments constrain what states become possible next.

Life therefore does more than occupy an environment. Life continuously constructs the material conditions of its own persistence.

But a material environment does not only stabilize what already exists. By changing constraint, it can also change biological possibility. A state that is physically possible may remain historically inaccessible until the system encounters the material sequence capable of reaching it.

That consequence moves anabiontics from the maintenance of biological state toward the navigation of biological state. It is the subject of Anabiontics II.